

Automated Laneway Home Feasibility Dashboard

Final Report Data 599

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Executive Summary

Problem Overview

Municipal zoning bylaws define critical rules for land use and development, including building height limits, setbacks, density, lot coverage, permitted uses, and accessory dwelling requirements. For laneway homes, these rules determine whether a structure is legally feasible on a specific parcel and what form it may take.

In practice, these rules are difficult to apply at scale. Bylaws are published as long PDF documents, tables, amendments, and zoning schedules with different terminology and update cycles across municipalities. Checking whether a parcel can support a laneway home is still largely manual: a practitioner must identify the parcel, confirm its zoning, locate the relevant sections, interpret the rules, and translate them into spatial constraints.

The project focuses on a practical prototype: a dashboard that displays city lot information through a GIS map, allows a user to select a lot, and shows whether a laneway home can be built on that lot — with traceable evidence for relevant rules. Burnaby, Vancouver, and Calgary are prioritized as the first prototype cities, supporting future generalization.

Motivation and Purpose

The purpose of this project is to convert zoning bylaw information into a structured and verifiable rule layer that supports parcel-level feasibility analysis. It connects rule extraction, normalization, verification, GIS-based spatial analysis, and dashboard presentation.

- Reduce manual effort by allowing users to inspect lot-level laneway home feasibility directly on a map.
- Improve consistency by applying normalized rules rather than relying on repeated manual interpretation.
- Maintain transparency by storing the source evidence for each extracted rule and making it accessible in the dashboard.

What We Built

We built an end-to-end automated pipeline with four stages: Extract — a large language model reads the bylaw PDF and pulls out candidate rules (height, setbacks, lot coverage) from both text and complex tables; Verify — a separate checking system approves only rules with direct evidence from the source document, flagging uncertain cases for human review; GIS — verified rules are spatially applied to parcel data to define buildable areas and fit building footprints for each lot; Visualise — results are displayed in an interactive Felt-based dashboard where a user selects any lot and instantly sees the applicable rules and building constraints.

Results

Verified rules and GIS-calculated parcel attributes are loaded directly into the dashboard. Any homeowner, developer, or planner can click on a lot and immediately see whether a laneway home is buildable, along with the applicable constraints and bylaw evidence — no prior knowledge of zoning documents required. Adding a new city requires only a bylaw PDF and a configuration file, making the pipeline a scalable foundation for future expansion across BC municipalities.

1. Introduction

1.1 Problem Overview

Municipal zoning bylaws define critical rules for land use and development, including building height limits, setbacks, density, lot coverage, permitted uses, and accessory dwelling requirements. For laneway homes and similar small-scale residential buildings, these rules determine whether a structure is legally feasible on a specific parcel and what form it may take.

In practice, these rules are difficult to apply at scale. Bylaws are published as long PDF documents, tables, amendments, and zoning schedules. Different municipalities use different terminology, document structure, rule exceptions, and update cycles. As a result, checking whether a parcel can support a laneway home is still largely manual: a practitioner must identify the parcel, confirm its zoning, locate the relevant bylaw sections, interpret the rules, and then translate them into spatial constraints.

The project scope focuses on a practical prototype: a dashboard that displays city lot information through a GIS map, allows a user to select a lot, and shows whether a laneway home can be built on that lot. If a laneway home is feasible, the dashboard should indicate the applicable building type or style and provide traceable evidence for the rules used in the decision. Due to the short project timeline, Burnaby is prioritized as the first prototype city, while early research on Surrey and Vancouver is used to support future generalization.

1.2 Motivation and Purpose

The purpose of this project is to convert zoning bylaw information into a structured and verifiable rule layer that can support parcel-level feasibility analysis in a dashboard. The project is not only a rule extraction task; it also connects rule extraction, rule normalization, verification, GIS-based spatial application, and dashboard presentation.

- Reduce manual effort by allowing users to inspect lot-level laneway home feasibility directly on a map.
- Improve consistency by applying normalized rules rather than relying on repeated manual interpretation.
- Maintain transparency by storing the source evidence for each extracted rule and making it accessible in the dashboard.
- Create a Burnaby-first prototype that can later be extended to additional municipalities with different bylaw formats.

2. Background Research

Prior work shows that land-use regulation can support computational urban analysis when rules are represented in structured formats. Waddell (2002) introduces UrbanSim as a framework for modeling urban development and demonstrates the value of structured land-use assumptions for scenario analysis and decision support. This supports the project goal of converting zoning constraints into data that can be applied programmatically.

Legal and regulatory text also presents challenges for automated extraction because it often contains conditional language, exceptions, cross-references, and domain-specific terminology. Niklaus et al. (2021) show that natural language processing methods can process complex legal language, but their work also highlights the need for careful evaluation when legal text is converted into structured predictions. For this project, that motivates the use of verification and human-review routing rather than treating model output as final.

Axelrod (2023) discusses the potential of machine learning for compiling standardized zoning data across jurisdictions and identifies major challenges such as fragmented documents, inconsistent formats, and varying local definitions. These issues are directly reflected in this project: Burnaby, Surrey, and Vancouver use different structures for laneway-related rules, so the system must normalize extracted rules into a shared representation before they can be used in GIS analysis.

3. Data Sources and Interesting Features

The project uses three main data categories: municipal bylaw PDFs, parcel geometry data, and zoning district data. The final prototype prioritizes Burnaby, while Surrey and Vancouver are reviewed during the early research stage to understand cross-city variation and to design a more general rule schema.

3.1 Zoning Bylaw PDFs

Zoning bylaw PDFs are the primary source for rules that determine laneway home feasibility. The zoning bylaw PDFs are obtained from official municipal government websites to ensure that all extracted rules are based on authoritative source documents. Relevant rule categories include permitted building type, minimum front/rear/side setbacks, building separation, lot coverage, height, floor area, parking-related requirements, and special exceptions. Burnaby is the main implementation target because the prototype needs a complete end-to-end result within the project timeline.

An important feature of the bylaw data is that critical rules may appear in both legal-text clauses and complex tables. In Burnaby, many setback and lot-coverage rules are concentrated in regulatory tables. This makes rule extraction more difficult because the system must preserve relationships between row labels, column headers, merged cells, repeated values, and conditional notes.

3.2 Parcel Geometry and Zoning District Data

Parcel geometry data provides the shape and size of individual lots. Zoning district data identifies the zoning category associated with each parcel. The GIS files are obtained from municipal Open Data Portals, including parcel geometry and

zoning district datasets used for spatial analysis. Together, these GIS layers allow the system to connect a selected lot to the applicable zoning rules and to calculate the buildable area created by setback requirements.

4. Aims and Objectives

4.1 Research Questions

1. How can laneway-home-related rules be extracted from municipal bylaw PDFs that contain both legal text and complex regulatory tables?
2. How can extracted rules be normalized into a consistent structure that supports GIS application and future cross-city generalization?
3. How can a verification layer check extracted rules against their cited evidence before they are used in a parcel-level dashboard?
4. How can setback rules be applied to parcel geometry so that the dashboard can show a lot-level buildable area and feasibility result?

5. Tools, Methodology, and Techniques

5.1 Extracting Laneway House Rules from City Bylaws

This section describes the method used to extract structured laneway house rules from municipal zoning bylaw PDFs. Municipal bylaws are long, heterogeneous, and legally nested documents. Rules relevant to laneway houses are rarely located in a single section. They may appear in use-permission schedules, district dimensional standards, parking sections, definitions, general development regulations, exception clauses, or regulatory tables. This makes laneway house rule extraction different from ordinary document question answering: the system must not only retrieve relevant text, but also determine which legal object the rule applies to, what constraint it imposes, and where the supporting evidence appears in the original bylaw.

The pipeline was developed to address limitations observed in many publicly available legal LLM and RAG-based tools. Most of these systems are designed around retrieving passages and generating answers from them. While useful for legal search or question answering, a purely RAG-based approach can miss relevant clauses, retrieve incomplete context, or apply a rule to the wrong object when the bylaw contains multiple building types, use categories, or nested exceptions. For zoning analysis, this is a serious limitation because a numeric value or permission status is only meaningful when it is tied to the correct object, such as a laneway house, dwelling unit, accessory building, parcel, yard, or parking requirement.

Therefore, this project develops an object-based rule extraction pipeline rather than a simple passage-retrieval system. The pipeline decomposes the bylaw into smaller evidence units, identifies rule-bearing content, groups related evidence into Graph/RAG packs, and then uses Gemini to extract structured rule objects from compact target-scoped inputs. Each extracted rule must connect a legal object to a specific constraint, operator, value, unit, approval status, condition, exception, and source reference. This design makes numeric values and key legal terms, such as permitted, prohibited, required, minimum, and maximum, explainable and traceable back to the original evidence.

The method is designed around three requirements. First, extraction must be object-aware: it should focus on laneway houses and equivalent local terms while retaining broader parent-category rules only when they clearly apply to the target object. Second, extraction must be evidence-traceable: each rule should be linked back to the original page, block, or table evidence so that its interpretation can be audited. Third, extraction must remain cost-efficient and scalable: the LLM receives selected evidence batches rather than unfiltered bylaw text, allowing the pipeline to process large bylaw documents economically while reducing the risk of missing or misapplying important rules.

5.1.1 Block Cutting and local candidate selection

The first stage converts each bylaw PDF into local text blocks. The PDF is parsed page by page, and extracted text is divided into smaller line-based blocks. Each block stores its source document, page number, local block ID, reading order, line range, and text content. This provides a finer unit of analysis than full pages or full documents, which is important because a single bylaw page can contain several unrelated regulations.

Local candidate selection is then applied to identify blocks that may contain laneway house rules. Each block is scored using three groups of terms: target terms, parent terms, and universal zoning terms. Target terms refer to the housing type of interest, such as “laneway house,” “laneway home,” “backyard suite,” “coach house,” or other city-specific equivalents. Parent terms capture broader categories that may apply to the target use, such as dwelling unit, residential building, accessory building, parcel, yard, floor area, parking, and access. Universal zoning terms capture common rule concepts such as setback, height, lot coverage, minimum, maximum, separation, permitted use, discretionary use, prohibition, and driveway access.

Blocks that directly mention target terms are treated as anchor blocks. The pipeline retrieves nearby blocks around these anchors to form candidate windows, so that headings, parent clauses, list items, and exceptions are not separated from the rule text they modify. Obvious low-value windows, such as index entries or bare use-list mentions without substantive regulatory language, are removed. The output of this stage is a high-recall set of candidate evidence blocks for downstream discovery and filtering.

5.1.2 RAG Structure Construction

After local candidate selection, the pipeline identifies which candidate blocks are likely to contain rule-bearing content. This stage does not rely only on exact target-term matches, because many relevant clauses apply through broader legal categories rather than directly naming laneway houses. Each candidate block is therefore annotated with regulatory, semantic, and structural signals, including target-term matches, parent-category matches, measurement expressions, cross-references, section context, list-item structure, and noise indicators.

Regulatory signals include modal and rule-bearing language such as must, shall, required, permitted, discretionary, prohibited, subject to, minimum, maximum, not exceed, setback, separation, parking, floor area, height, density, coverage, and permit. Numeric expressions with units such as metres, square metres, percentages, storeys, dwelling units, or parking stalls are also treated as strong rule evidence. These signals help distinguish substantive regulations from headings, page headers, amendment notes, table-of-contents entries, or other non-rule text.

The discovery stage also assigns selected blocks to functional rule lanes, such as use permission, dimensional standard, parking, permit process, cross-reference, definition, target context, and related context. These lanes are used to organize evidence before extraction. For example, use-permission evidence is handled separately from dimensional standards, and definitions or cross-references can be retained as context without being treated as final measurable constraints.

A key purpose of this stage is to preserve nested legal structure. Municipal bylaws often express rules as a parent clause followed by lettered or numbered child items. A child item may not repeat the target term, but it may still inherit meaning from the parent clause. The pipeline therefore keeps related list items together when they appear to share the same governing clause or section scope. This reduces the risk of extracting isolated fragments without their legal context.

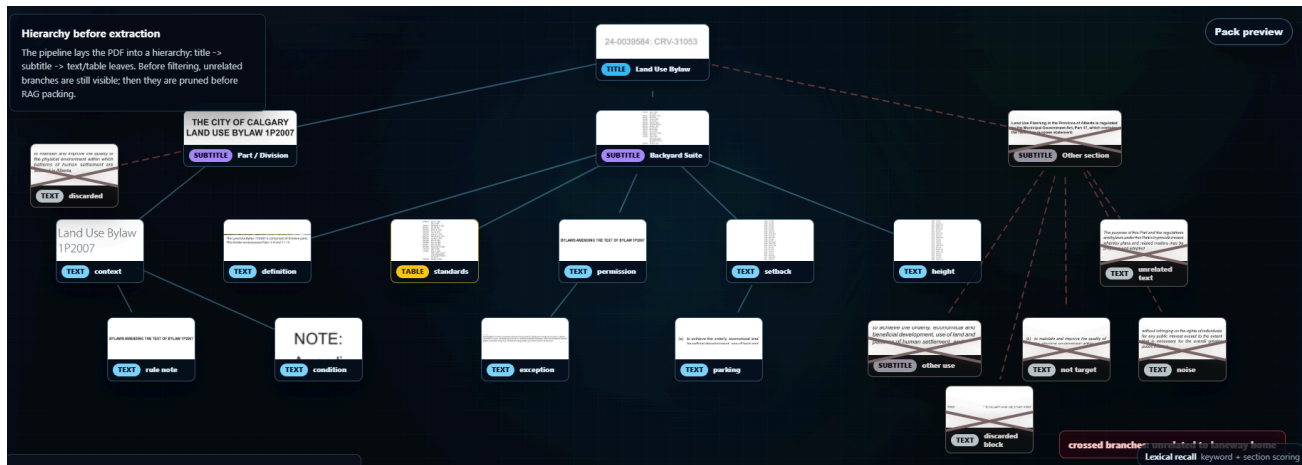


Figure 1: demonstration of RAG structure of a section of the PDF file, the grey block are the ones not related to laneway home bylaws

5.1.3 Lexical and BGE-M3 Based Block Classification and Packing

After rule discovery, the pipeline further filters and ranks the selected evidence using lexical scoring and BGE-M3 semantic classification. This stage reduces the amount of text sent to the LLM while retaining both directly relevant laneway house rules and indirectly applicable parent-category rules.

Lexical scoring provides an interpretable first layer. It identifies blocks that explicitly contain target terms, parent-category terms, zoning rule keywords, or measurement language. Because lexical matches are easy to inspect, this layer also supports auditing: selected evidence can be explained by the terms and regulatory signals that caused it to be retained.

BGE-M3 semantic classification is used to improve recall when bylaws express applicable rules without using exact target terminology. For example, a laneway house rule may be embedded under broader language such as “all dwelling units,” “accessory residential buildings,” “residential buildings,” “all parcels,” or “all uses in the district.” Semantic similarity helps retain these blocks even when they would be missed by exact keyword matching alone.

The final retention decision combines semantic relevance with structural evidence from earlier stages. Blocks are prioritized when they occur near a target anchor, belong to the same section as a target rule, contain measurable standards, fall within a relevant rule lane, or inherit meaning from a parent clause. This keeps the extraction input compact while preserving the legal scope needed for accurate rule interpretation.



Figure 2: Graph-based RAG representation of bylaw evidence. The graph organizes filtered PDF blocks into rule-specific evidence packs before LLM extraction. The central node represents the city bylaw, while connected nodes represent RAG packs and supporting source blocks. Colors indicate rule categories such as core target rules, use permissions, dimensional standards, table regions, and contextual evidence. This structure preserves regulatory context and reduces unnecessary token usage.

5.1.4 Graph/RAG Evidence Pack Construction

The retained evidence blocks are organized into Graph/RAG evidence packs before extraction. Each pack groups related blocks by source page, section scope, and rule role. The main pack types include core target rules, use permissions, district dimensional standards, universal applicable rules, and context-only material.

This pack structure prevents unrelated legal content from being mixed in the same extraction input. Use-permission clauses are separated from dimensional standards; district-specific requirements are separated from city-wide or parent-category rules; and context-only evidence is retained without being treated as a final rule by itself. This organization helps the LLM interpret each evidence group according to its regulatory function.

The packs also preserve source traceability. Each evidence item keeps its original source ID, page reference, selection reason, and contextual role. Repeated or near-duplicate evidence is grouped before extraction so that the model does not repeatedly process the same rule text. The result is a compact set of evidence packs that preserves legal context while reducing unnecessary API input.



Figure 3: The packages are grouped by their rule types and sent to online API for official rule retrieval parallelly.

5.1.5 Target-Scoped Evidence Adaptation

Before LLM extraction, the Graph/RAG packs are converted into target-scoped evidence inputs. This step narrows each pack to the text most relevant to the target building type while preserving the clauses needed to interpret it.

If a block directly contains a target term, the pipeline keeps the target-relevant clause and nearby governing language. If a block does not directly mention the target, it is retained only when its scope clearly applies through a parent category, such as dwelling units, residential buildings, accessory buildings, parcels, yards, parking, or district-wide dimensional standards. Broad context and unrelated use-list items are removed at this stage to reduce the chance that the model extracts rules for non-target uses.

Table evidence is prepared separately. When regulatory tables are available as visual crops, the table images are retained for table-specific extraction. This is important because OCR text often loses row-column relationships, while the image crop preserves the layout needed to interpret dimensional standards, conditions, and footnotes.

5.1.6 Gemini-Based Rule Extraction

The extraction stage uses Gemini to convert target-scoped evidence into structured rule records. Text evidence and table evidence are processed through separate extraction paths because they require different interpretation strategies.

For text evidence, Gemini receives compact batches of RAG evidence blocks. The prompt asks the model to extract atomic zoning rules only. Each rule should represent one independently testable constraint, permission, prohibition, exception, measurement rule, or approval requirement. The model is instructed to cite the source ID, preserve inherited scope, split compound clauses into separate rules, retain exception and approval language, and skip headings or context-only text.

For table evidence, Gemini receives regulatory table images together with table metadata. The table prompt asks the model to preserve row hierarchy, column hierarchy, units, conditions, exceptions, ranges, and visible footnote markers. This is necessary because many zoning standards are meaningful only when a cell value is interpreted together with its row label, column heading, and section scope.

The extraction output is structured rather than narrative. Each rule contains fields for source reference, rule object, constraint type, subject, operator, value, unit, condition, exception, evidence text, and warnings. Operators and units are normalized

where possible, so that minimum requirements, maximum requirements, fixed values, permissions, prohibitions, and measurement units can be represented consistently in the final rule table.

5.1.7 Post-Processing, Deduplication, and Review Routing

The final stage consolidates raw extracted rules into a cleaner rule table. Text-derived and table-derived rules are merged into a common schema, duplicate records are folded, and fields such as units, operators, rule objects, and approval types are standardized.

Deduplication is necessary because the same regulation may appear in multiple forms, such as a text clause, a table cell, a repeated district section, or OCR text from a table image. The pipeline compares normalized rule attributes, including source reference, rule object, subject, operator, value, unit, condition, and evidence text. When overlapping records are detected, the pipeline keeps the clearest or most specific representation while preserving source traceability.

The post-processing stage also separates clean rules from rules that require review. Rules are routed to the review queue when they contain unresolved exceptions, incomplete evidence, source warnings, operator-direction conflicts, unit-object conflicts, non-numeric measurement values, or ambiguous cross-references. This prevents uncertain records from being silently accepted while keeping them available for manual inspection.

The final output is a structured rule table suitable for downstream GIS analysis, along with audit files and a review queue. The clean rule table can be used to compare parcel attributes against extracted zoning requirements such as use permission, setbacks, building height, floor area, lot coverage, parking, and access conditions.

5.2 Verification Layer

After extraction, the candidate rules are not sent directly to the GIS map. They first enter a separate verification layer that checks whether the extracted value, unit, operator, rule type, source page, and scope are actually supported by the cited bylaw evidence.

This layer was built as a deterministic Python verifier rather than another LLM judge. That design kept the final decision auditable and low-cost: the extraction pipeline could be improved and rerun, while the same candidates could be checked repeatedly without paying for a new model call for every verification decision.

The verifier uses a conservative rule: *extraction proposes, verification proves, and GIS consumes verified-only outputs*. If the evidence is incomplete, ambiguous, outside the supported rule families, or separated from its legal context, the candidate is held in review instead of being promoted.

rule_object	constraint_type	operator	value	unit	condition	source_stream
permitted_use	permitted_use	permitted			Principal Use	gemini_table_image
lot_width	minimum	>=	5	m	Interior Lot	gemini_table_image
parking_stall_width	minimum_dimension	>=	2.50	m	90-degree parking; dwelling units	gemini_table_image
floor_space_ratio	computation_method	required			based on pre-subdivision site area	gemini_text_batch

Table 1: Extracted Rule Format Demonstration (Shortened for Readability)

In addition to the public-facing map, the project includes a reviewer-oriented dashboard for communicating verification status. This view is designed for project partners and reviewers: it shows which product run is current, how many rules were verified or held for review, whether false verified rules remain at zero, and why only verified outputs should flow into downstream GIS layers.

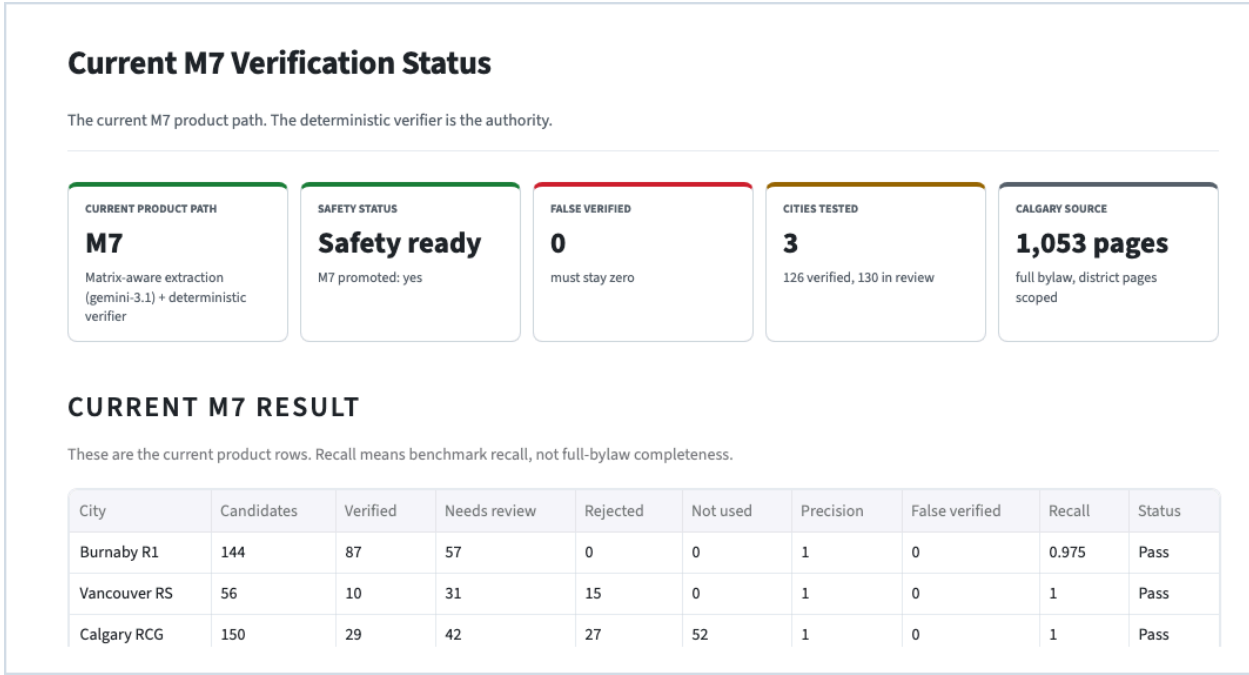


Figure 4: : M7 reviewer dashboard summarizing the verified-only safety status and per-city verification results.

5.3 GIS Analysis

Determining whether a laneway house can be legally built on a lot is a geospatial problem. All geospatial analysis was completed using Python scripts in combination with QGIS. One Python script finds buildable areas where laneway houses can legally be built, and another determines if a sample building footprint fits within those buildable areas. These are generic scripts that can run for multiple municipalities, including Burnaby and Calgary, despite differences in GIS schema. To implement the solution for a municipality, only a list of parameters needs to be provided that outlines extracted rules for that municipality along with schema details.

5.3.1 Labelling Lot Boundaries

The first step to obtaining buildable area is labelling each lot boundary as front, side, or rear. Though most lots are simple rectangles, there are many irregular shapes as well. A generic algorithm was developed that can label lot boundaries for lots of any shape, outlined in Figure 5.

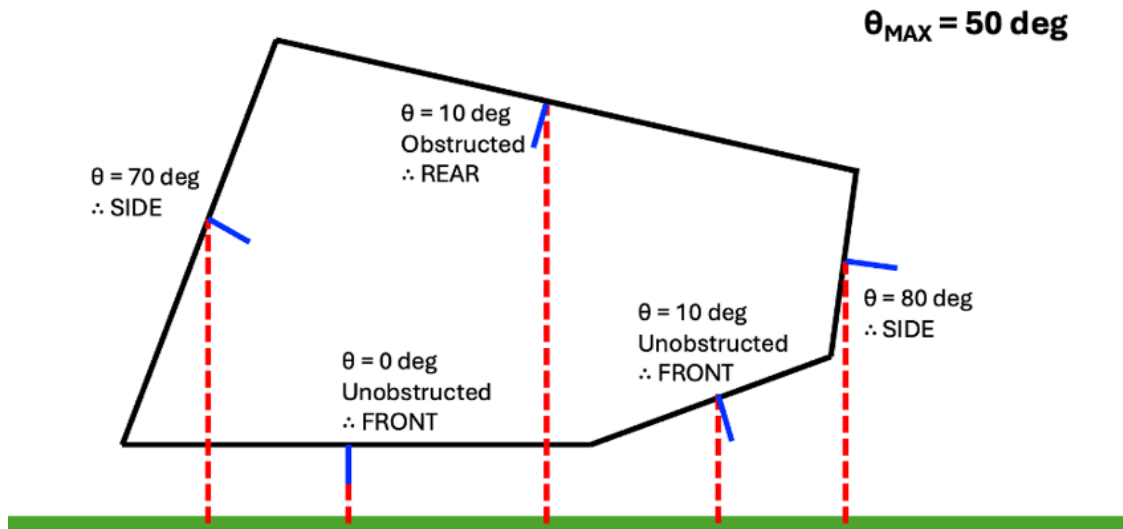


Figure 5: Diagram to outline the lot boundary labelling algorithm. Bee-lines (red) extend from the lot boundary midpoints to the street (green). The angle, θ , between a bee-line and the perpendicular bisector (blue) plays an important role in labelling.

First, the algorithm finds every lot boundary's bee-line, the shortest line between the boundary's midpoint and its street. Then, it finds the acute angle between each bee-line and the boundary's perpendicular bisector. This angle, θ , represents how much the boundary faces its street. If θ is below θ_{MAX} , then it is either a front or rear boundary. If the bee-line does not intersect any other boundary, then it is a front boundary. If it intersects a boundary from the same lot, then it is a rear

boundary. Otherwise, it is a side boundary. A value of $\theta_{MAX} = 50^\circ$ is used so that boundaries with $\theta \approx 45^\circ$ are labelled consistently even with minor deviations in angle.

5.3.2 Applying Setbacks

Setback rules determine how far a laneway house needs to be from different lot boundaries. The particular setback rule that is applied depends on the boundary's label and whether the boundary borders a neighbouring lot, street, or lane. By subtracting the illegal spaces to build along the boundaries, the remaining space is the buildable area if there were no existing structures.

Boundary Label	Boundary Type	Setback Applied
Front	Street	Street Non-Flanking
Side	Neighbouring lot	Interior Side
	Street	Street Flanking
	Lane	Interior Side
Rear	Neighbouring lot	Interior Rear
	Street	Street Flanking
	Lane	Lane

Table 2: Conditions under which different setbacks are applied.

5.3.3 Subtracting Existing Structures

For the purpose of developing laneway houses, only principal buildings need to be considered. Accessory buildings such as detached garages and sheds can be demolished. The largest building on each lot is labelled as principal, others are labelled as accessory. Principal buildings are subtracted from buildable areas with a buffer to account for the minimum distance allowed between front and rear principal buildings. The remaining space is the true buildable area where a laneway house can legally be built.

5.3.4 Fitting a Laneway House Footprint

To determine if a building's footprint fits in a lot's buildable area (Figure 6), many positions and rotations are tested. First, starting positions are placed in a grid spaced every 4 m that fall inside the lot. From each starting position, a greedy search repeatedly moves the footprint in whichever of eight directions that reduces the footprint area beyond the buildable area most. The translational step begins at 0.5 m and halves every time no direction improves the fit. The search stops once the footprint either becomes fully contained or can no longer reduce its area beyond the buildable area. Each starting position is tested using every possible footprint rotation in 10 deg steps until containment is found. Since the scope of the project is limited to laneway houses, placement is only allowed if the footprint's centroid is closer to the lot's average rear centroid than its average front centroid. In other words, laneway houses must be closer to the rear than the front.

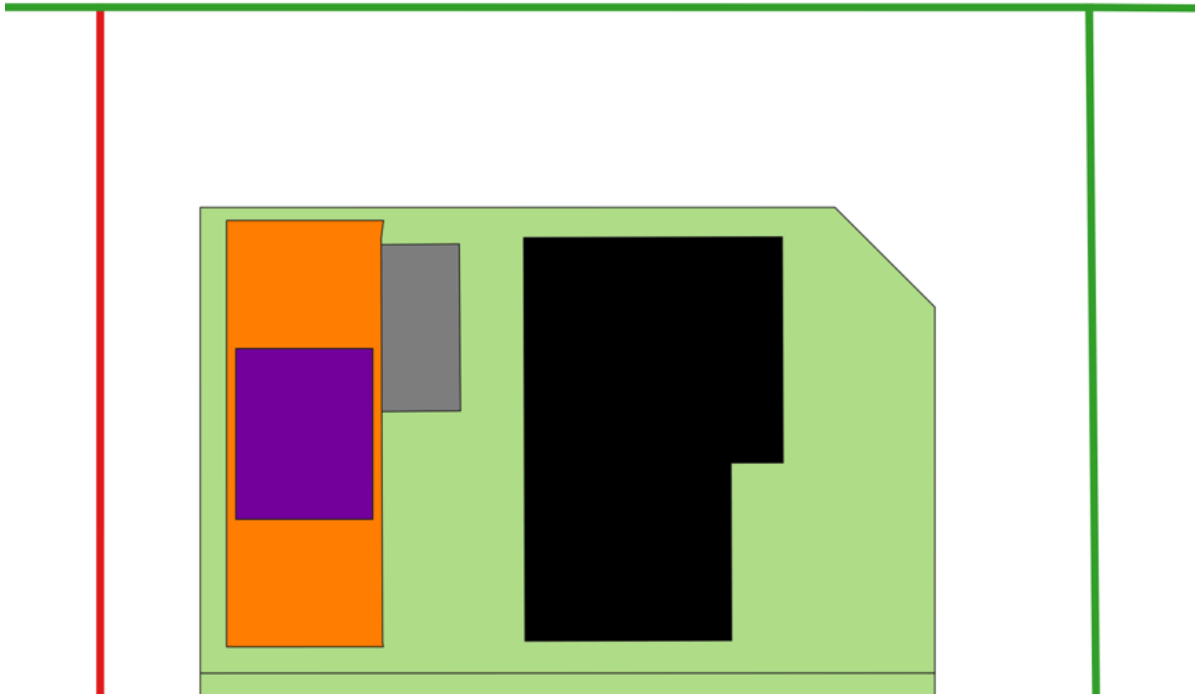


Figure 6: Lot (light green) with a fitted footprint (purple) within the buildable area (orange). Buildable area accounts for the principal building (black), but ignores the accessory building (grey). The lot borders two streets (dark green) and a lane (red).

5.4 Map Dashboard Visualization

The rule extraction and GIS analysis pipelines produce structured outputs — verified rules and buildable area geometries — but these outputs are not directly accessible to end users. A visualization layer was built to connect these outputs to an interactive map that any user can explore without technical knowledge. The dashboard is built in Felt, a browser-based GIS platform that supports GeoJSON uploads, multiple map layers, and configurable hover tooltips. No software installation is required for users.

5.4.1 Rule-to-Map Pipeline

Verified rules are stored in a file called `gis_rule_contract.json`, which is produced by the verification pipeline. However, this file cannot be uploaded to a map directly — rules only describe what is allowed in a zone, but they have no location information. To show rules on a map, they must first be connected to the parcel boundaries they apply to.

Two scripts handle this process. The first script reads the verified rules file and converts each rule into a row in a spreadsheet, recording the rule type, limit, unit, condition, and bylaw reference. The second script reads the parcel map file, looks up which zone each parcel belongs to, and attaches the matching rules to each parcel. The final output is a single map file where every parcel already contains its own rules as data fields. This file is uploaded to Felt as the main map layer. Four additional

layers are then added on top: buildable areas, fitted footprints, existing buildings, and lot boundary lines. This process was completed for both Burnaby (R1 zone, 36,915 parcels) and Calgary (RCG zone, 1,156 parcels).

5.4.2 Dashboard Layers and User Experience

The final dashboard for each city contains five layers. The parcel layer is the primary interactive layer: clicking any parcel opens a popup showing the lot address, zone, area, parcel type, and the filtered rule tooltip. The buildable area layer shows the zone within each lot where a laneway home can legally be placed after setbacks are applied. The fitted footprint layer shows a sample building footprint placed inside the buildable area. The buildings layer shows existing structures on the lot. The lot lines layer provides boundary labels for orientation. Together, these layers allow a user to answer three questions from a single map interaction:

- What rules apply to this lot?
- Where on the lot can a laneway home be built?
- Does a standard footprint fit within that space?



Figure 7: Burnaby interactive dashboard showing a selected R1 parcel with zone code, area, parcel type (laned, corner), and filtered zoning rules displayed in the hover tooltip.

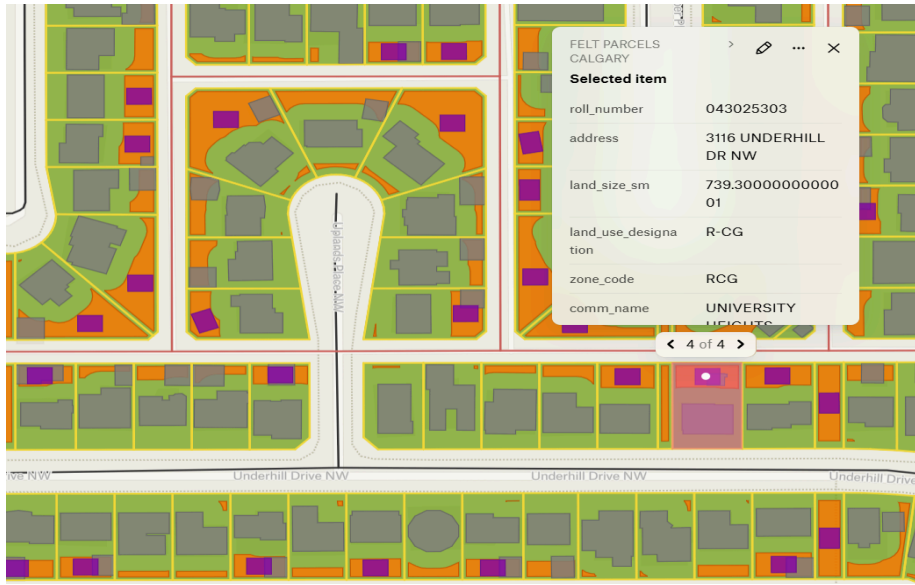


Figure 8: Calgary interactive dashboard showing a selected RCG parcel with lot details including address, land size, land use designation, and zone code displayed in the hover tooltip.

6. Analysis and Interpretation

6.1 Rule Extraction Result

The official pipeline was tested on three cities and now shows stable generalization across different bylaw formats: Burnaby uses compact table-heavy zoning pages, Calgary uses a very large bylaw, and Vancouver is mostly text-based. The same pipeline structure can process all three without city-specific extraction logic.

City	Target Type	Runtime	API Calls	Token Use	Est. Cost	Clean Rules	Review Queue	Coverage
Burnaby	Small-scale multi-unit / rowhouse	165.0s	7 text + 4 table	~101k	~CA\$0.5	66	0	complete
Calgary	Backyard suite	92.3s	26 text + 1 table	~226k	~CA\$0.4–0.6	143	7	complete
Vancouver	Laneway house	46.6s	7 text + 0 table	~62k	~CA\$0.1	44	9	complete

Table 3: Overall Summary of Rule Extraction Results

The current pipeline can complete rule extraction for all three cities in under three minutes. This is a major improvement over the early Calgary experiment, where sending large portions of the 1,053-page bylaw directly to the API took around 15 minutes and cost about CA\$8. The improvement comes from avoiding full-document extraction. Instead, the pipeline first filters and compresses PDF blocks locally, builds Graph/RAG evidence packs, separates text and table evidence, and only sends target-relevant batches to the API. This keeps token use low while preserving the evidence most important for rule modelling.

In the current evaluation, the pipeline captured all target-relevant numeric evidence, including dimensional standards such as height, setbacks, floor area, lot coverage, parking quantities, and other measurable requirements. It also captured approximately 95% of target-relevant non-numeric evidence, such as use permissions, approval conditions, exceptions, and general applicability clauses. The remaining missed items were mostly non-numeric clauses that were either not directly relevant to the target building type or only weakly connected through broad contextual language. Overall, the pipeline is now fast, low-cost, and reasonably generalized across different city formats, target building types, and both text-based and table-based rule sources.

6.2 Verified Rule Results and Safety Interpretation

The extraction table above reports structured candidate rules. The verification table below reports the stricter safe-to-map set: candidates whose value, unit, operator, source, and scope are supported by cited bylaw evidence. Lower verified counts are expected because ambiguous, duplicated, or unsupported candidates stay in review instead of being promoted.

For the final M7 product run, verified precision was 1.00 across Burnaby, Vancouver, and Calgary, with 0 false verified rules and 0 false approvals.

Verification contract: extraction and RAG may be uncertain; the verifier cannot be. Any unresolved candidate stays in review until stronger evidence or human review closes the gap.

City	Checked	Verified	Review	Rejected / Unused	Precision / FV
Burnaby R1	144	87	57	0 / 0	1.00 / 0
Vancouver RS	56	10	31	15 / 0	1.00 / 0
Calgary R-CG	150	29	42	27 / 52	1.00 / 0

Table 4: Verified-only M7 Results Used by the Dashboard and GIS Contract

The review counts are intentional and should not be read as missing rules. They show that the system is fail-closed: it prefers human review over guessing. This is especially important for table-heavy bylaws, conditional clauses, and large sources such as Calgary's 1,053-page bylaw, where nearby text may contain values that look correct but refer to a different rule.

The final validation also included 485 unit tests and an adversarial benchmark with poisoned rule cases and poisoned evidence bundles. These tests check known failure patterns such as unsupported values, flipped operators, wrong rule families, table-column mixups, and conditional clauses being treated as unconditional.

6.3 GIS Analysis Result

Upon visual inspection, the GIS solutions to define buildable areas and fit footprints are extremely accurate and robust to irregularly shaped lots. Unfortunately, there are no GIS tests to calculate accuracy. With more time and resources, a professional surveyor could be hired to manually identify the correct setback to apply to lot boundaries across a subset of lots in different municipalities. Then, the result of the scripts can be compared with the surveyor's verdict to calculate the percentage of lot boundaries that apply the correct setback. Similarly, buildings could be manually labelled to calculate the accuracy of that component of the algorithm.

Despite lacking GIS tests, cul-de-sacs were noticed to sometimes have front boundaries mislabelled as sides. This is because streets are represented as centrelines instead of polygons, so front bee-lines can be erroneously oblique, as seen in Figure 9. This could be addressed by developing an algorithm to convert street centrelines to representative polygons.

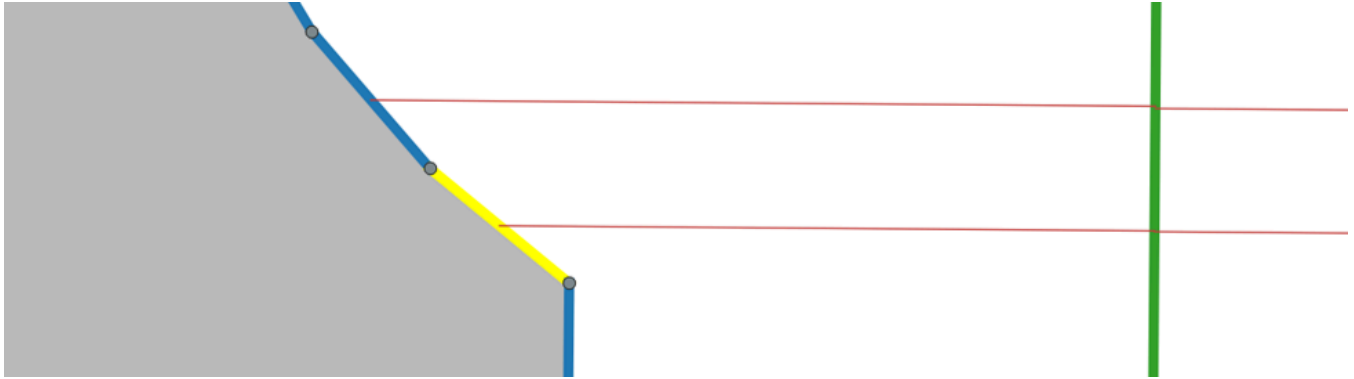


Figure 9: Zoomed-in on a lot in a cul-de-sac (grey). Bee-lines (red) extend from the lot boundary midpoints to the street centreline (green). The yellow boundary is erroneously identified as a side instead of a front because it doesn't sufficiently face the street's centreline, even though it does in fact face the street.

Mislabelled lot boundaries can also occur when a single street wraps around a lot. In these cases, the closest point from a rear boundary midpoint to the street can be at such a great angle that the rear is misidentified as a side, as seen in Figure 10.

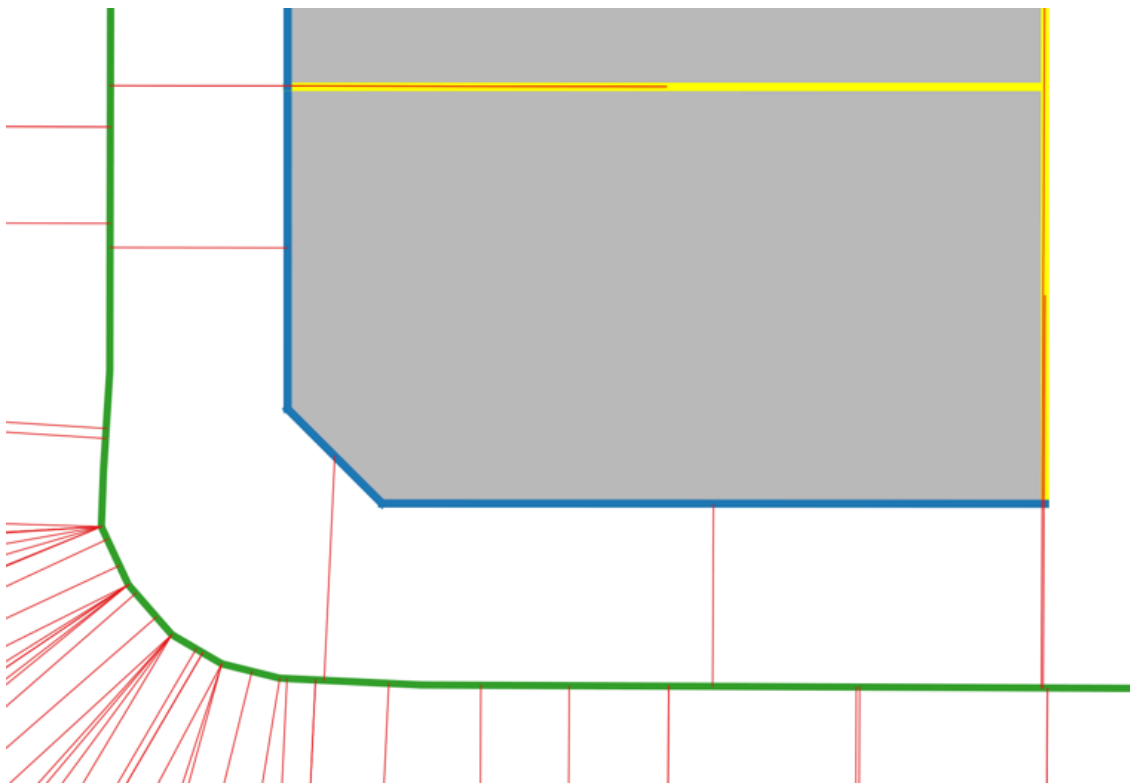


Figure 10: Lot (grey) with a rear boundary mislabelled as a side on the righthand side due to a wrapping street. Bee-lines (red) extend from the lot boundary midpoints to the street (green).

6.4 Map Dashboard Result

The dashboard successfully connects verified zoning rules to individual parcels and displays them in an accessible format. Clicking any parcel on the map produces a tooltip showing only the rules that apply to that specific lot, organized by category. This confirms that the rule-to-map pipeline works end-to-end for both Burnaby and Calgary.

A key strength of the dashboard is its accessibility to users with no technical background. Rather than requiring knowledge of GIS software or zoning terminology, a user only needs to find their property on the map and click on it. The relevant rules appear immediately in plain language, organized by category, and filtered to reflect the specific characteristics of that lot. This removes the need to manually read through bylaw documents, cross-reference zone schedules, or interpret conditional rule language.

The dashboard currently displays only rules from the verified set. Rules that remain in the human review queue are intentionally excluded, which means some relevant rules may not appear for a given parcel. This is a conservative but correct behaviour — showing unverified rules could mislead users. As more rules move from the review queue to verified, the tooltip will automatically become more complete without any changes to the dashboard itself.

7. Conclusion

7.1 Rule Extraction

The rule extraction pipeline was able to extract zoning rules from Burnaby, Calgary, and Vancouver with good coverage and acceptable cost. The same workflow was used across all three municipalities, showing that the pipeline has a reasonable level of generalization across different bylaw formats and target housing types. The extracted outputs include use permissions, dimensional standards, setbacks, height limits, lot coverage, parking-related rules, and table-based regulations needed for downstream GIS analysis.

A major improvement of the final pipeline is that it does not send the full bylaw document to the API. Instead, the PDF is divided into blocks, filtered, compressed, and organized into a Graph/RAG structure before extraction. This structure groups related evidence blocks by page, section, rule type, and target relevance, which helps preserve context while removing unrelated content. As a result, the LLM receives smaller and more focused extraction packs, reducing token use and improving extraction consistency. This design greatly reduced runtime and cost. For example, an early Calgary experiment on the 1,053-page bylaw took around 15 minutes and cost approximately CA\$8, while the current pipeline can complete extraction for each tested city in under 3 minutes at a much lower cost.

The pipeline also separates text rules from table-based rules. This is important because many zoning bylaws store dimensional standards in tables rather than paragraphs. By sending table crops through a dedicated table extraction branch, the pipeline can preserve rules that are difficult to recover reliably from OCR text alone.

One limitation is that the current workflow does not include web scraping. The relevant bylaw PDFs still need to be found and downloaded manually before rule extraction can begin. In future work, an upstream document discovery and update-monitoring module could be added to make the system more automated.

7.2 Rule Verification

After extraction, the rules are passed into the deterministic verification layer. This means the extraction stage is not the final authority on rule correctness. Its role is to produce structured, source-linked candidates quickly and cheaply, while the verifier checks exact evidence support before any rule can enter the GIS contract or dashboard. This separation also reduces cost: LLM calls are used for candidate extraction and review assistance, while final promotion is handled by auditable Python logic. In short, the lower verified count is a safety result, not a loss of useful extraction work.

7.3 GIS Analysis

The GIS algorithms used to generate buildable areas and fit footprints are extremely accurate and robust to irregularly shaped lots based on a visual inspection. However, boundaries are sometimes mislabelled in cul-de-sacs and when a street wraps around a lot. There were insufficient resources to develop GIS tests to calculate accuracy. However, this could be done in the future with the help of a professional surveyor to obtain a reference with guaranteed fidelity. With tests, parameters in the algorithm such as θ_{MAX} could be fine-tuned to maximize accuracy.

When subtracting existing structures from buildable areas, the current algorithm does not distinguish between existing laneway houses and accessory buildings. Both are ignored when defining buildable areas. This is a problem since housing developers do not typically tear down one laneway house to build another. If there was more time, methods to distinguish between existing laneway houses and accessory buildings would have been explored. It may be impossible to reliably determine from footprint geometry and position alone.

Although the GIS scripts can support the schemas for Burnaby and Calgary, there is not yet a generic solution to support the schema of any potential municipality. Having conditions in the scripts themselves to handle different schemas, as is done currently, is not a scalable solution. Ideally, an automated solution can be developed to convert required GIS files to a standardized schema to allow for simple scalable scripts.

7.4 Map Dashboard

For the interactive dashboard, Felt was chosen as the visualization platform because it requires no software installation, supports direct GeoJSON and GeoPackage uploads, and allows hover tooltips to display parcel-level rule information. A key advantage of the dashboard is that a user who already knows their property address can search for it directly on the map and immediately see all applicable zoning rules for that specific lot, replacing a process that would otherwise require manually reading through hundreds of pages of bylaw documents.

However, Felt has limitations for this use case. The tooltip display is plain text only, with no ability to format tables or highlight key values. Felt also does not support user input, so a user cannot enter a custom footprint size and see whether it fits within the buildable area of a selected lot.

For a first prototype, Felt is a practical and effective choice for demonstrating that verified zoning rules can be connected to a parcel map and made accessible to any user. A more capable future dashboard could allow users to input footprint dimensions and receive an immediate feasibility result for their specific lot, making the tool more directly actionable for homeowners and developers.

8. References

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